7.5 Sorting a Positional List

In Section 5.5.2, we introduced the ***insertion-sort*** algorithm, in the context of an array-based sequence. In this section, we develop an implementation that operates on a PositionalList, relying on the same high-level algorithm in which each element is placed relative to a growing collection of previously sorted elements.

We maintain a variable named marker that represents the rightmost position of the currently sorted portion of a list. During each pass, we consider the position just past the marker as the pivot and consider where the pivot's element belongs relative to the sorted portion; we use another variable, named walk, to move leftward from the marker, as long as there remains a preceding element with value larger than the pivot's. A typical configuration of these variables is diagrammed in Figure 7.15. A Python implementation of this strategy is given in Code 7.17.

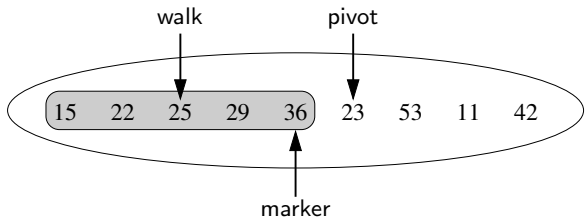


Figure 7.15: Overview of one step of our insertion-sort algorithm. The shaded elements, those up to and including marker, have already been sorted. In this step, the pivot's element should be relocated immediately before the walk position.

```
1  def insertion_sort(L):
2    """Sort PositionalList of comparable elements into nondecreasing order."""
3    if len(L) > 1:                          # otherwise, no need to sort it
4      marker = L.first( )
5      while marker != L.last( ):
6        pivot = L.after(marker)             # next item to place
7        value = pivot.element( )
8        if value > marker.element( ):       # pivot is already sorted
9          marker = pivot                    # pivot becomes new marker
10       else:                               # must relocate pivot
11         walk = marker                     # find leftmost item greater than value
12         while walk != L.first( ) and L.before(walk).element( ) > value:
13           walk = L.before(walk)
14         L.delete(pivot)
15         L.add_before(walk, value)         # reinsert value before walk
```

Code Fragment 7.17: Python code for performing insertion-sort on a positional list.